第六章 近独立粒子的构成的系统.

§1

系统：

能级：$\varepsilon_1, \varepsilon_2, \cdots \varepsilon_l, \cdots$

简并度：$\omega_1, \omega_2 \cdots \omega_l \cdots$

粒子数：$a_1, a_2 \cdots a_l \cdots$

给出约束：$N = \sum a_l$，$E = \sum a_l \varepsilon_l$

三种分布的状态数：

1. Boltzman：$\Omega_{MB} = \frac{N!}{\prod a_l!} \prod \omega_l^{a_l}$

2. Bose-Einstein：$\Omega_{BE} = \prod \frac{(\omega_l + a_l - 1)!}{a_l!(\omega_l - 1)!}$

ω_l 选 a_l 个，可重复 → a_l 个量，$\omega_l - 1$ 个格.

3. Fermi-Dirac：$\Omega_{FD} = \prod \frac{\omega_l!}{a_l!(\omega_l - a_l)!}$

经典极限：$\frac{a_l}{\omega_l} \ll 1$，$\frac{\Omega_{MB}}{N!} = \Omega_{BE} = \Omega_{FD}$（全同性原理）

§2 分布的推导

等几率原理 & 最概然分布. ← $S = k \ln \Omega$，为 S 取最大分布，$\ln\Omega$ 取最小分布.

Stirling 公式：$\ln(n!) \approx \int_1^n \ln x \, dx \approx n(\ln n - 1)$

约束：$N = \sum a_l$，$E = \sum a_l \varepsilon_l$

1. Boltzman：

$$\ln \Omega = \ln N! + \ln \prod \omega_l^{a_l} - \ln \prod a_l! \quad (a_l \gg 1)$$

$$\approx N(\ln N - 1) + \sum a_l \ln \omega_l - \sum \ln a_l!$$

$$\approx N \ln N + \sum a_l [\ln \omega_l - \ln a_l]$$

$$\delta \ln \Omega = \sum \left(\ln \frac{\omega_l}{a_l}\right) \delta a_l = 0 \quad (\omega_l \text{ 不可变})$$

考虑 Lagrange 函数：

$$\delta \ln \Omega - \alpha \delta N - \beta \delta E = \sum \left\{ \ln\left(\frac{w_i}{a_i}\right) - 1 - \beta \varepsilon_i \right\} \delta a_i = 0$$

$$\Rightarrow \ln\left(\frac{w_i}{a_i}\right) - \alpha - \beta \varepsilon_i = 0 \rightarrow a_i = w_i e^{-\alpha - \beta \varepsilon_i}$$

$$* \quad \delta^2 \ln \Omega = \sum \frac{\partial}{\partial w_i}\left(-\frac{w_i}{a_i^2}\right) \delta^2 a_i = -\sum \frac{\delta^2 a_i}{a_i} < 0 \quad (a_i > 0)$$

最概然 <-> 实际分布 了

$$\ln \frac{\Omega + \delta \Omega}{\Omega} \approx -\frac{1}{2} \sum \frac{\delta^2 a_i}{a_i} \ll 1$$

2. Bose - Einstein

$$\ln \Omega = \sum \ln (w_i + a_i - 1)! - \sum \ln (w_i - 1)! - \sum \ln a_i!$$

$$\approx \sum (w_i + a_i) \left\{ \ln (w_i + a_i) - 1 \right\} - \sum w_i (\ln w_i - 1) - \sum a_i (\ln a_i - 1)$$

$$= \sum (w_i + a_i) \ln (w_i + a_i) - \sum w_i \ln w_i - \sum a_i \ln a_i$$

$$\delta \ln \Omega = \sum \left\{ \ln (w_i + a_i) - \ln a_i \right\} \delta a_i$$

$$\delta L = \sum \left\{ \ln \left(\frac{w_i + a_i}{a_i}\right) - \alpha - \beta \varepsilon_i \right\} \delta a_i = 0$$

$$\Rightarrow \frac{w_i + a_i}{a_i} = e^{\alpha + \beta \varepsilon_i} \rightarrow a_i = \frac{w}{e^{\alpha + \beta \varepsilon_i} - 1}$$

3. Fermi - Dirac

$$\ln \Omega = \sum w_i (\ln w_i - 1) - \sum (w_i - a_i) \left\{ \ln (w_i - a_i) - 1 \right\} - \sum a_i (\ln a_i - 1)$$

$$= \sum w_i \ln w_i - \sum (w_i - a_i) \ln (w_i - a_i) - \sum a_i \ln a_i$$

$$\delta \ln \Omega = \left\{ \ln (w_i - a_i) - \ln a_i \right\} \delta a_i$$

$$\delta L = \left\{ \ln \left(\frac{w_i - a_i}{a_i}\right) - \alpha - \beta \varepsilon_i \right\} \delta a_i = 0$$

$$\Rightarrow \frac{w_i - a_i}{a_i} = e^{\alpha + \beta \varepsilon_i} \rightarrow a_i = \frac{w_i}{e^{\alpha + \beta \varepsilon_i} + 1}$$

超导混合态的统计关联效应

§3 热力学量的统计表达式

1. U、N

$$U = \sum a_i \varepsilon_i = \sum \omega_i \varepsilon_i e^{-\alpha-\beta\varepsilon_i}$$

$$N = \sum a_i = \sum \omega_i e^{-\alpha-\beta\varepsilon_i}$$

引入配分函数 $Z = \sum \omega_i e^{-\beta\varepsilon_i}$

$$\Rightarrow N = e^{-\alpha} Z$$

$$U = -N \frac{\partial}{\partial \beta} \ln Z$$

2. W、Q.

广义力 $Y = \sum \frac{\partial \varepsilon_i}{\partial y} a_i = \sum \frac{\partial \varepsilon_i}{\partial y} \omega_i e^{-\alpha-\beta\varepsilon_i}$

$$= \frac{N}{Z} \left(-\frac{1}{\beta}\right) \frac{\partial Z}{\partial y} = -\frac{1}{\beta} \frac{\partial \ln Z}{\partial y}$$

压强 $P = -\frac{\partial E}{\partial V} = P(T, V)$，物态方程.

$$\overline{DQ} = dU - \overline{DW}$$

$$= \sum a(\omega_i \varepsilon_i) - dy \cdot \sum \frac{\partial \varepsilon_i}{\partial y} a_i$$

$$= \sum \varepsilon_i \, da_i$$

3. 二能级系统

$\varepsilon_1 = -\varepsilon$, $\varepsilon_2 = \varepsilon$, $\omega_1 = \omega_2 = 1$

对定域的：Σ 中粒子彼此可区分

配分函数 $Z = \sum \omega_i e^{-\beta\varepsilon_i} = e^{\beta\varepsilon} + e^{-\beta\varepsilon}$

$$e^{-\alpha} = \frac{N}{e^{\beta\varepsilon} + e^{-\beta\varepsilon}}$$

平均分布：$\bar{a}_1 = e^{-\alpha+\beta\varepsilon} = N \cdot \frac{e^{\beta\varepsilon}}{e^{\beta\varepsilon} + e^{-\beta\varepsilon}}$

$$\bar{a}_2 = e^{-\alpha-\beta\varepsilon} = N \cdot \frac{e^{-\beta\varepsilon}}{e^{\beta\varepsilon} + e^{-\beta\varepsilon}}$$

$T \to 0$, $\bar{a}_1 = N$, $\bar{a}_2 = 0$, $U \to -\varepsilon$

$T \to \infty$, $\bar{a}_1 = \bar{a}_2 = \frac{N}{2}$

$$\frac{U}{N} = \frac{3}{2}kT$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_{V,N_i} = \frac{3}{2}Nk$$

$$\frac{C_V}{Nk} = \frac{3}{2}$$

理想气体定容比热

$$T \to 0 : C \to 0$$

$$T \to \infty : C \to \frac{3}{2}Nk$$

C_V

T

4. S

$$dU = TdS - pdV + \mu dN$$

$$\left(\frac{\partial U}{\partial S}\right)_{V,N} = T \qquad \left(\frac{\partial U}{\partial V}\right)_{S,N} = -p \qquad \left(\frac{\partial U}{\partial N}\right)_{S,V} = \mu$$

$$dS = \frac{1}{T}dU + \frac{p}{T}dV - \frac{\mu}{T}dN$$

$$\left(\frac{\partial S}{\partial U}\right)_{V,N} = \frac{1}{T} \qquad \left(\frac{\partial S}{\partial V}\right)_{U,N} = \frac{p}{T} \qquad \left(\frac{\partial S}{\partial N}\right)_{U,V} = -\frac{\mu}{T}$$

$$\left(\frac{\partial S}{\partial U}\right)_{V,N}\left(\frac{\partial U}{\partial S}\right)_{V,N} = 1$$

$$\left(\frac{\partial S}{\partial V}\right)_{U,N}\left(\frac{\partial V}{\partial U}\right)_{S,N}\left(\frac{\partial U}{\partial S}\right)_{V,N} = -1$$

$$\left(\frac{\partial S}{\partial V}\right)_{U,N} = -\left(\frac{\partial U}{\partial V}\right)_{S,N}\left(\frac{\partial S}{\partial U}\right)_{V,N} = \frac{p}{T}$$

$$\left(\frac{\partial S}{\partial N}\right)_{U,V}\left(\frac{\partial N}{\partial U}\right)_{S,V}\left(\frac{\partial U}{\partial S}\right)_{V,N} = -1$$

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$$\left(\frac{\partial S}{\partial N}\right)_{U,V} = -\left(\frac{\partial U}{\partial N}\right)_{S,V}\left(\frac{\partial S}{\partial U}\right)_{V,N} = -\frac{\mu}{T}$$

$\Leftrightarrow \quad S = k \ln \Omega$

$= \frac{\partial \ln Z}{\partial \beta} \cdot \frac{\partial \beta}{\partial \varepsilon} \cdot Z = \frac{\partial \ln Z}{\partial \varepsilon} Z$

5. 又，

$F = U - TS = -N \frac{\partial \ln Z}{\partial \beta} - \frac{1}{\beta} (\ln Z^N - \ln N!)$

$= -NkT \ln Z$

6.

在经典极限条件下 $\Omega_{BE} = \Omega_{FD} = \frac{\Omega_{MB}}{N!}$

$S = k \ln \left(\frac{\Omega_{MB}}{N!}\right) = Nk \left(\ln Z - \beta \frac{\partial \ln Z}{\partial \beta}\right) - k \ln N!$

$F = -NkT \ln Z + kT \ln N!$

对二能级系统：

$T \to 0$ 时，$a_1 = N, a_2 = 0 \cdot \Omega = \frac{N!}{a_1! a_2!} = 1$

$S = 0.$

$T \to \infty$ 时，$\Omega = \frac{N!}{\left[\left(\frac{N}{2}\right)!\right]^2}$，$S = k \ln \frac{N!}{\left[\left(\frac{N}{2}\right)!\right]^2}$

7. 系数 α 的含义

化学势 $\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V}$

经典极限下，$F = -NkT \ln Z + kT \ln N!$

$\approx -kT \left[N \ln Z - N(\ln N - 1)\right]$

$= -NkT \ln \left(\frac{Z}{N} e\right)$

$\left(\frac{\partial F}{\partial N}\right)_{T,V} = -kT \left[\ln \left(\frac{Z}{N} e\right) + N \cdot \frac{N}{Ze} \cdot \left(-\frac{Ze}{N^2}\right)\right]$

$= kT \ln \left(\frac{Z}{N}\right) = -\alpha kT$

$\Rightarrow \quad \alpha = -\frac{\mu}{kT}$

8. 理想气体

$P = \frac{N}{\beta} \frac{\partial \ln Z}{\partial V}$，关键在于求出配分函数 Z

对单个粒子，

$\varepsilon = \frac{1}{2m} (p_x^2 + p_y^2 + p_z^2) \quad p_i = \frac{2\pi \hbar}{L} n_i$

$\varepsilon_n = \frac{\pi^2 \hbar^2}{2mL^2} n^2$

h: 양자 상태 하나가 차지하는 위상 공간 부피 $\sim h^3$ (h: Planck 상수)

$$Z = \sum_{n=0}^{\infty} e^{-\beta \varepsilon_n} = \int_{\text{phase space}} \frac{dq_1 \cdots dq_N\, dp_1 \cdots dp_N}{h^{3N}}\, e^{-\beta \mathcal{H}(q,p)}$$

자유 입자 기체의 분배함수:

$$Z = \int \frac{1}{h^{3N}} dx_1 dy_1 dz_1 \cdots dx_N dy_N dz_N\, dp_{x_1} dp_{y_1} dp_{z_1} \cdots dp_{z_N}\, e^{-\frac{1}{2mkT}(p_{x_1}^2 + p_{y_1}^2 + p_{z_1}^2 + \cdots)}$$

$$= \frac{V^N}{h^{3N}} \left(\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mkT}} dp \right)^{3N} \qquad \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$= \frac{V^N}{h^{3N}} (2\pi mkT)^{3N/2}$$

이상 기체 상태 방정식:

$$Z = \int_0^{\infty} e^{-\frac{1}{2mkT} p^2} \cdot \frac{V^N}{h^{3N}} (4\pi p^2)^N dp$$

$$= \frac{(4\pi)^N V^N}{h^{3N}} \int_0^{\infty} e^{-\frac{1}{2mkT} p^2} p^{2N} dp \qquad \int_0^{\infty} e^{-\alpha x^2} x^n dx = \frac{\Gamma\left(\frac{n+1}{2}\right)}{2\alpha^{\frac{n+1}{2}}}$$

$$= \frac{(4\pi)^N V^N}{h^{3N}} \cdot \Gamma\left(N + \tfrac{1}{2}\right) \left(\frac{1}{2mkT}\right)^{N+\frac{1}{2}} = \frac{V^N}{h^{3N}} (2\pi mkT)^{3N/2}$$

$$\Rightarrow P = \frac{N}{\beta} \cdot \frac{\partial}{\partial V} \ln Z = \frac{N}{\beta} \frac{1}{V} = \frac{NkT}{V}$$

$$PV = NkT$$

$$U = -\frac{N}{Z} \frac{\partial Z}{\partial \beta} = -\frac{N}{Z} \left(-\frac{3}{2}\right) \cdot \frac{Z}{\beta} = \frac{3}{2} NkT$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{3}{2} Nk$$

$$Z = -NkT \ln\left(\frac{N!}{e^N}\right)$$

$$= -NkT \left[\ln N! - N + \frac{3}{2} \ln(2\pi mkT) + \ln\left(\frac{V}{h^3}\right) - \ln N \right]$$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V$$

$$= Nk \left[\frac{5}{2} + \frac{3}{2} \ln(2\pi mkT) + \frac{3}{2} \ln\left(\frac{2\pi m}{h^3}\right) + \ln\left(\frac{V}{N}\right) \right]$$

9. 고전적 극한에 대한 조건: $e^{\alpha} \gg 1$

$$\alpha = -\frac{\mu}{kT} \rightarrow \mu \ll -kT$$

$$e^{\alpha} = \frac{N}{Z} = \frac{N}{V} \cdot \left(\frac{h^2}{2\pi mkT}\right)^{3/2} \gg 1, \quad T \uparrow, V \uparrow, n \downarrow$$

또 다른 해석 방법:

드브로이 파장 $\lambda_D = \frac{h}{\sqrt{2\pi mkT}} \rightarrow n\lambda_D^3 \ll 1$

二 粒子无穷远 非简并

10. Maxwell 速度分布

在体积 V 内，

$$N = \int \frac{V}{h_0^3} dp_x dp_y dp_z \cdot e^{-\alpha - \frac{1}{2mkT}(p_x^2 + p_y^2 + p_z^2)}$$

$$\Rightarrow e^{-\alpha} = \frac{N}{V} \cdot \left(\frac{h_0^2}{2\pi mkT}\right)^{3/2}$$

$$f(v_x, v_y, v_z)\, dv_x dv_y dv_z$$

$$= n \cdot \left(\frac{h_0^2}{2\pi mkT}\right)^{3/2} e^{-\frac{m}{2kT}(v_x^2 + v_y^2 + v_z^2)} dv_x dv_y dv_z$$

$$f(v)\, dv$$

$$= n \cdot \left(\frac{h_0^2}{2\pi mkT}\right)^{3/2} e^{-\frac{mv^2}{2kT}} 4\pi v^2 dv$$

$$\frac{df(v)}{dv}\Big|_{v=v_m} = 0 \Rightarrow v_m = \sqrt{\frac{2kT}{m}}$$

$$\bar{v} = \sqrt{\frac{8kT}{\pi m}}$$

$$v_s = \sqrt{\int_0^\infty v^2 f(v)\, dv} = \sqrt{\frac{3kT}{m}}$$

撞壁次数 了

$$df \cdot dA\, dt = f(v_x, v_y, v_z)\, dv_x dv_y dv_z \cdot v_x dA\, dt$$

$$\Gamma = \int f \cdot v_x\, dv_x dv_y dv_z$$

$$= n \cdot \sqrt{\frac{kT}{2\pi m}} = \frac{1}{4} n\bar{v}$$

11. 能均分定理

对于处在温度为 T 的平衡状态的经典系统，粒子能量中每一个平方项的均值为 $\frac{1}{2}kT$

e.g. 对动能 $\varepsilon_p = \frac{1}{2}\sum c_i p_i^2$

$$\overline{\frac{1}{2}c_j p_j^2} = \frac{1}{N}\int \frac{1}{2}c_j p_j^2 e^{-\alpha-\beta\varepsilon}\cdot dq_1\sim dq_r\,dp_1\sim dp_r \Big/ h_0^r$$

$$= \frac{1}{Z}\int \frac{1}{2}c_j p_j^2 e^{-\beta\varepsilon}\,dq_1\sim dq_r\,dp_1\sim dp_r \Big/ h_0^r$$

$$= \frac{1}{Z}\int \frac{1}{2}c_j p_j^2 e^{-\frac{\beta}{2}c_j p_j^2}\,dp_j\cdot\int e^{-\beta\varepsilon'}\,dq_1\sim dq_r\,dp_1\sim dp_r \Big/ h_0^r$$

$$= \frac{1}{Z}\left(-\frac{1}{2\beta}p_j\cdot e^{-\frac{\beta}{2}c_j p_j^2}\Big|_{-\infty}^{+\infty} + \frac{1}{2\beta}\int e^{-\frac{\beta}{2}c_j p_j^2}dp_j\right)$$

$$\int e^{-\beta\varepsilon'}\,dq_1\sim dq_r\,dp_1\sim dp_r \Big/ h_0^r$$

$$= \frac{1}{2\beta Z}\int e^{-\beta\varepsilon}\cdot dq\cdots dp\cdots \Big/ h_0^r$$

$$= \frac{1}{2\beta Z}\cdot Z = \frac{1}{2}kT$$

应用：① 单原子分子　　忽略了电子对热容的影响。

$$\varepsilon = \frac{1}{2m}(p_x^2+p_y^2+p_z^2)，\text{因为 } kT \ll eV$$

$$\bar{\varepsilon} = \frac{3}{2}kT，\quad U = \frac{3}{2}NkT.\quad C_V = \frac{3}{2}Nk,\quad \gamma = \frac{5}{3}$$

② 双原子分子：

$$\varepsilon = \frac{1}{2M}(p_x^2+p_y^2+p_z^2) + \frac{1}{2I}\left(p_\theta^2 + \frac{1}{\sin^2\theta}p_\varphi^2\right)$$

$$+ \frac{1}{2\mu}p_r^2 + U(r)$$　　不考虑振动自由度，常温不激发。

$$\bar{\varepsilon} = \frac{5}{2}kT$$

③ 固体

$$\varepsilon = \frac{1}{2m}p^2 + \frac{1}{2}m\omega^2 q^2$$

$$\bar{\varepsilon} = 3kT$$

(2) 理想气体的内能和热容

$\varepsilon = \varepsilon_t + \varepsilon_r + \varepsilon_v$ 平动 + 振动 + 转动

$z = z_t \cdot z_r \cdot z_v = \sum_{t,r,v} e^{-\beta(\varepsilon_t+\varepsilon_r+\varepsilon_v)}$

$= z_t \cdot z_r \cdot z_v$

$U = -N \frac{\partial}{\partial \beta} \ln z = -N \frac{\partial}{\partial \beta} (\ln z_t + \ln z_r + \ln z_v)$

$= U_t + U_r + U_v$

$C = C_t + C_r + C_v$

(1) 平动

$U_t = \frac{3}{2} N k T$, $C_t = \frac{3}{2} N k$

(2) 振动 ~ 线性谐振子

$\varepsilon_v = (n + \frac{1}{2}) \hbar\omega$

$z_v = \sum_{n=0}^{\infty} e^{-\beta(n+\frac{1}{2})\hbar\omega}$

$= e^{-\frac{1}{2}\beta\hbar\omega} \frac{1}{1-e^{-\beta\hbar\omega}} = \frac{e^{\frac{1}{2}\beta\hbar\omega}}{e^{\beta\hbar\omega}-1}$

$U_v = -N \cdot \frac{\partial}{\partial \beta} \ln z_v = N\hbar\omega \left(\frac{1}{2} + \frac{e^{-\beta\hbar\omega}}{1-e^{-\beta\hbar\omega}}\right)$

$C_v = \left(\frac{\partial U_v}{\partial T}\right)_V = N\hbar\omega \frac{e^{\beta\hbar\omega}-\hbar\omega}{(e^{\beta\hbar\omega}-1)^2} \cdot \frac{1}{kT^2}$

$= Nk \left(\frac{\hbar\omega}{kT}\right)^2 \frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega}-1)^2}$

引入特征温度: $k\Theta_v = \hbar\omega$

$C_v = Nk \left(\frac{\Theta_v}{T}\right)^2 \frac{e^{\Theta_v/T}}{(e^{\Theta_v/T}-1)^2}$ 一般 $\Theta_v \gg T$.

(3) 转动

$\varepsilon_r = \frac{\hbar^2 l(l+1)}{2I}$

$z_r = \sum_l (2l+1) e^{-\beta \frac{\hbar^2 l(l+1)}{2I}}$ $k\Theta_r = \frac{\hbar^2}{2I}$

$= \int_0^{\infty} (2l+1)\, e^{-\frac{\hbar^2 l(l+1)}{2I k_B T}}$ $\quad \theta_r \ll T$

$= \int_0^{\infty} (2l+1)\, e^{-\frac{\hbar^2 l(l+1)}{2I k_B T}}\, dl$

$= \frac{2I k_B T}{\hbar^2}\, e^{-\frac{\hbar^2 l(l+1)}{2I k_B T}} \Big|_0^{\infty} = \frac{T}{\theta_r} = T \cdot \frac{2k}{\hbar^2 \beta} = \frac{2k}{\beta\hbar^2}$

$U^r = -N \cdot \frac{d}{d\beta} \ln Z^r = NkT$

$C^r = Nk$

3. 固体热容理论.

1. Einstein 理论. (完成不了)

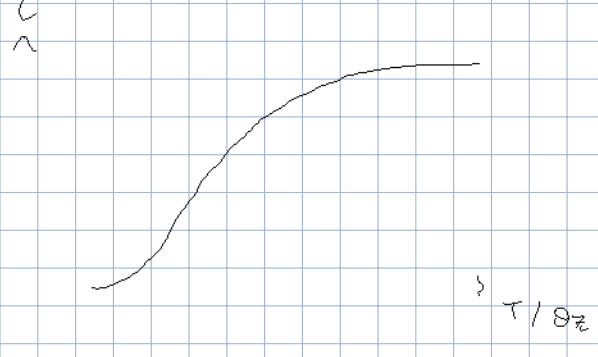
固体: $3N$ 个振子, $\varepsilon = (n+\frac{1}{2})\hbar\omega$

$Z = \sum e^{-\beta(n+\frac{1}{2})\hbar\omega} = \frac{e^{\frac{1}{2}\beta\hbar\omega}}{e^{\beta\hbar\omega}-1}$

$U = -3N \cdot \frac{\partial}{\partial\beta} \ln Z = 3N\hbar\omega\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega}-1}\right)$

$C = \left(\frac{\partial U}{\partial T}\right)_V = 3Nk \left(\frac{\hbar\omega}{kT}\right)^2 \cdot \frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega}-1)^2}$

$= 3Nk \left(\frac{\theta_E}{T}\right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T}-1)^2} \approx 3Nk \left(\frac{\theta_E}{T}\right)^2 e^{-\theta_E/T}$ ($T \ll \theta_E$ 时)



2. Debye 理论.

设第 i 个自由度对应坐标为 ξ_i:

系统的势能: $\Phi = \Phi_0 + \sum \left(\frac{\partial \Phi}{\partial \xi_i}\right)_0 \xi_i + \frac{1}{2} \sum \left(\frac{\partial^2 \Phi}{\partial \xi_i \partial \xi_j}\right)_0 \xi_i \xi_j$ (平衡, $=0$)

$= \sum \frac{p_i^2}{2m} + \sum \frac{1}{2} a_{ij} \xi_i \xi_j + \Phi_0$

$\xi_i \to q_i$ 对角化, ζ

$$= \frac{1}{2}\sum_{i=1}^{3N}\left(p_i^2 + \omega_i^2 q_i^2\right) + \phi_0$$

$$= \sum_{i=1}^{3N}\hbar\omega_i\left(n_i + \frac{1}{2}\right) + \phi_0$$

$$Z = e^{-\beta\phi} = e^{-\beta\left[\sum_{i=1}^{3N}\hbar\omega_i\left(n_i+\frac{1}{2}\right) + \phi_0\right]}$$

$$= e^{-\beta\phi_0}\cdot\prod_{i=1}^{3N}\sum_{n_i=0}^{\infty} e^{-\beta\hbar\omega_i\left(n_i+\frac{1}{2}\right)}$$

$$= e^{-\beta\phi_0}\cdot\prod_{i=1}^{3N}\frac{e^{-\frac{1}{2}\beta\hbar\omega_i}}{e^{\beta\hbar\omega_i}-1}$$

$$u = -\frac{\partial}{\partial\beta}\ln Z = \phi_0 + \sum_{i=1}^{3N}\hbar\omega_i\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega_i}-1}\right)$$

$$= u_0 + \sum_{i=1}^{3N}\frac{\hbar\omega_i}{e^{\beta\hbar\omega_i}-1}$$

结合能 + 振动能.

如何求频谱 $\{\omega_i\}$?

$$D(\omega)\,d\omega = \frac{V}{2\pi^2}\left(\frac{1}{C_l^3} + \frac{1}{C_t^3}\right)\omega^2\,d\omega = B\omega^2\,d\omega$$

$$\int_0^{\omega_D} D(\omega)\,d\omega = B\cdot\int_0^{\omega_D}\omega^2\,d\omega = \frac{B}{3}\omega_D^3 = 3N$$

$$\Rightarrow \omega_D^3 = \frac{9N}{B}$$

$$u = u_0 + \int_0^{\omega_D} D(\omega)\cdot\frac{\hbar\omega}{e^{\beta\hbar\omega}-1}\,d\omega$$

$$= u_0 + B\int_0^{\omega_D}\frac{\hbar\omega^3}{e^{\beta\hbar\omega}-1}\,d\omega \qquad \Theta_D = \frac{\hbar\omega_D}{k}$$

$$= u_0 + B\left(\frac{kT}{\hbar}\right)^4\cdot\int_0^{\Theta_D/T}\frac{\left(\frac{\hbar\omega}{kT}\right)^3}{e^{\frac{\hbar\omega}{kT}}-1}\,d\left(\frac{\hbar\omega}{kT}\right)$$

$$= u_0 + 9NkT\cdot\left(\frac{T}{\Theta_D}\right)^3\cdot\int_0^{\Theta_D/T}\frac{\left(\frac{\hbar\omega}{kT}\right)^3}{e^{\frac{\hbar\omega}{kT}}-1}\,d\left(\frac{\hbar\omega}{kT}\right)$$

① 高温, $T \gg \Theta_D$

$$u \approx u_0 + 9NkT\cdot\left(\frac{T}{\Theta_D}\right)^3\cdot\frac{1}{3}\cdot\left(\frac{\Theta_D}{T}\right)^3 = u_0 + 3NkT.$$

② 低温

$$u = u_0 + 3N\cdot\frac{\pi^4}{5}\cdot\frac{T^4}{\Theta_D^3},$$

$$C_V = 3Nk \left(\frac{4\pi^4}{5}\right) \cdot \left(\frac{T}{\Theta_D}\right)^3 \propto T^3$$